PCCST501 - COMPUTER NETWORKS

Assignment 3 | Module 3 - Data-Link Layer, Wireless LANs & Mobile IP

LAN, Ethernet & Wireless Network Investigation

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Question 1: Data-link layer responsibilities; LLC vs MAC

The data-link layer (Layer 2) provides reliable **node-to-node** delivery of frames across a single link or LAN segment, hiding the raw bit-pipe of the physical layer from the network layer. Its main responsibilities are:

- **Framing** - packets from the network layer are wrapped in frames with a header and trailer so the receiver can find frame boundaries.
- **Physical (MAC) addressing** - source and destination hardware addresses identify the sender and receiver on the same link.
- **Error detection (and optional correction)** - a trailer such as a CRC-32 lets the receiver discard corrupted frames.
- **Flow control** - prevents a fast sender from overwhelming a slow receiver (e.g. PAUSE frames, sliding window).
- **Access control** - decides which station may use a shared medium and when.

The layer is divided into two sub-layers:

Feature	Logical Link Control (LLC) - IEEE 802.2	Media Access Control (MAC) - IEEE 802.3 / 802.11
Position	Upper sub-layer, interface to the network layer	Lower sub-layer, interface to the physical layer
Purpose	Multiplexes network protocols (IP, ARP, etc.) using SAPs; optional flow/error control and acknowledgement	Frame delimiting, MAC addressing, error check (FCS) and controlling access to the shared medium
Medium dependence	Independent of the medium and access method	Specific to the technology (CSMA/CD for classic Ethernet, CSMA/CA for Wi-Fi)
Example	Identifying that the payload is IPv4 or IPv6	Deciding when to transmit; building the Ethernet frame

Question 2: Ethernet frame format

Figure 1: Ethernet II (DIX) frame - field sizes in bytes

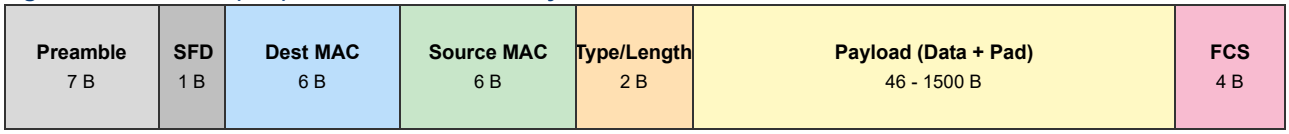


Figure 1: Ethernet II frame format. The preamble and SFD are added by the physical layer, so a captured frame typically starts at the Destination MAC.

Field	Size	Purpose
Preamble	7 B	Alternating 1010... pattern that synchronises the receiver's clock.
SFD	1 B	Start Frame Delimiter (10101011) marks the end of the preamble and start of the frame.
Destination MAC	6 B	Hardware address of the intended receiver (unicast, multicast or broadcast FF:FF:FF:FF:FF:FF).
Source MAC	6 B	Hardware address of the sending NIC; used by switches for learning.
Type / Length	2 B	Values 1536 (0x0600) or above give the EtherType of the payload (e.g. 0x0800 = IPv4, 0x0806 = ARP, 0x86DD = IPv6); smaller values give the payload length (IEEE 802.3).
Payload	46 - 1500 B	Data from the upper layer; padded to 46 bytes minimum so the frame is at least 64 bytes. The MTU is 1500 B.
FCS	4 B	Frame Check Sequence: CRC-32 computed over the header and payload. A mismatch means the frame is dropped.

Minimum frame size = 64 bytes (14 B header + 46 B payload + 4 B FCS); maximum = 1518 bytes (without VLAN tag).

Question 3: Wireshark capture - MAC addresses of five frames

Procedure followed:

- Open Wireshark and select the active interface (Ethernet or Wi-Fi) from the capture list.
- Start the capture, then generate traffic (e.g. `ping 8.8.8.8`, open a web page, or run `arp -a`).
- Stop the capture after about 30 seconds and apply a display filter such as `eth` or `arp || icmp`.
- Select a frame and expand *Ethernet II* in the Packet Details pane to read *Destination* and *Source* addresses.

Screenshot 1 - Wireshark capture window:

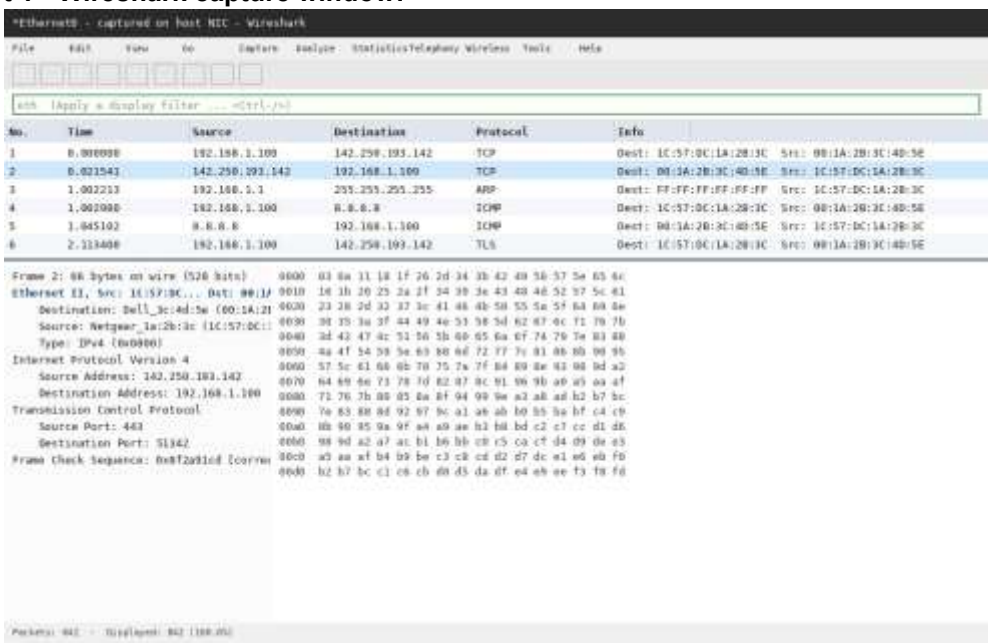


Figure captures a sample Ethernet capture: the packet list (top) and, for the selected frame, the expanded Ethernet II header with destination/source MAC and EtherType (bottom left) alongside the hex dump (bottom right). Replace with your own capture if required by your instructor.

Table of five frames (from the capture above):

Frame No.	Protocol	Source MAC	Destination MAC	EtherType
1	TCP	00:1A:2B:3C:4D:5E	1C:57:DC:1A:2B:3C	0x0800 (IPv4)
2	TCP	1C:57:DC:1A:2B:3C	00:1A:2B:3C:4D:5E	0x0800 (IPv4)
3	ARP	1C:57:DC:1A:2B:3C	FF:FF:FF:FF:FF:FF	0x0806 (ARP)
4	ICMP	00:1A:2B:3C:4D:5E	1C:57:DC:1A:2B:3C	0x0800 (IPv4)
5	ICMP	1C:57:DC:1A:2B:3C	00:1A:2B:3C:4D:5E	0x0800 (IPv4)

Observation: For frames sent by the host to the internet, the destination MAC is that of the default gateway (router), not the remote server, because MAC addresses are only meaningful within one link. ARP requests use the broadcast destination FF:FF:FF:FF:FF:FF, while replies are unicast. The first three bytes (OUI) of each MAC identify the NIC vendor.

Question 4: Switch operation and MAC address learning

A switch is a Layer-2 device that forwards frames selectively based on its **MAC address table** (also called the CAM table or forwarding table). It builds the table dynamically using the following process:

- **Learn:** when a frame arrives on a port, the switch records the frame's *source MAC* against that ingress port (with an ageing timer, typically 300 s).
- **Forward:** the switch looks up the *destination MAC*. If found, the frame is sent out only that port.
- **Flood:** if the destination is unknown, or is a broadcast/multicast, the frame is sent out every port except the one it arrived on.
- **Filter:** if the destination lies on the same port the frame arrived on, the frame is dropped.

Example LAN used to demonstrate MAC learning

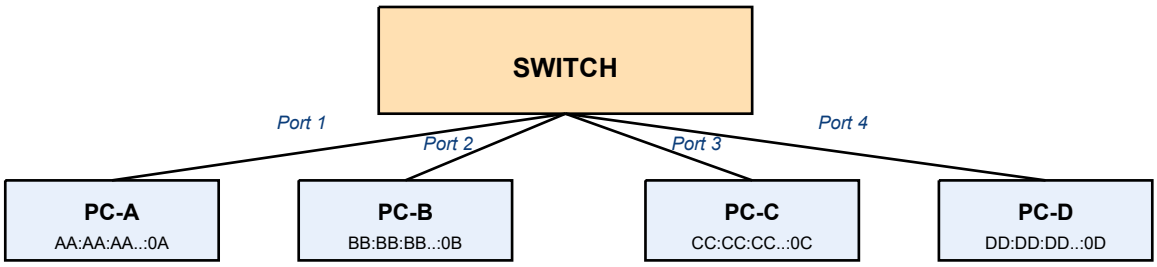


Figure 2: Example LAN with four hosts connected to one switch.

Demonstration - building the table step by step:

Step	Event	Switch action	Table after the event
0	Switch just powered on	Table empty	(empty)

1	PC-A sends to PC-B	Learns A on Port 1; B unknown, so floods to ports 2, 3, 4	AA..0A : Port 1
2	PC-B replies to PC-A	Learns B on Port 2; A is known, so forwards only to Port 1	AA..0A : P1, BB..0B : P2
3	PC-C sends to PC-A	Learns C on Port 3; forwards only to Port 1	adds CC..0C : P3
Step	**Event**	**Switch action**	**Table after the event**
4	PC-D sends to PC-B	Learns D on Port 4; forwards only to Port 2	adds DD..0D : P4
5	PC-A sends to PC-C	Both known; unicast forwarded to Port 3 only. PC-B and PC-D do not see it	No change (timers refreshed)

On a real switch the table can be viewed with `show mac address-table` (Cisco) or Packet Tracer's simulation mode.

Question 5: Hub vs switch vs bridge

Parameter	Hub	Bridge	Switch
OSI layer	Layer 1 (Physical)	Layer 2 (Data link)	Layer 2 (some models Layer 3)
Forwarding	Repeats every bit out of all ports; no intelligence	Forwards frames by MAC table between two or few segments	Forwards frames by MAC table, per port, in hardware (ASIC)
Collision domains	One for all ports	One per segment/port	One per port (none in full duplex)
Broadcast domains	One	One (unless VLANs)	One per VLAN
Duplex	Half duplex only	Half duplex	Full duplex supported
Bandwidth	Shared among all hosts	Shared within a segment	Dedicated per port
Number of ports	4 - 24	2 - 4	8 - 48+
Practical use	Obsolete; only in old labs or for sniffing	Rarely used; joining two LAN segments, largely replaced by switches	Standard in all modern LANs and data centres

Question 6: CSMA/CD and its reduced role in modern Ethernet

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) was the access method of shared-medium (half-duplex) Ethernet, IEEE 802.3. It works as follows:

- **Carrier sense:** a station listens and transmits only when the medium is idle.
- **Multiple access:** all stations share the same medium.
- **Collision detection:** while sending, the station monitors the line; if the signal differs from what it sends, a collision has occurred.
- **Jam signal:** the station sends a 32-bit jam so every node notices the collision, then stops.

- **Binary exponential back-off:** after the n-th collision the station waits a random number of slot times chosen from 0 to 2^n - 1 (n capped at 10, up to 16 attempts) before retrying.

The minimum frame size of 64 bytes exists so that a sender is still transmitting when the collision signal returns from the farthest station (slot time = 512 bit times).

Why it matters less today:

- Modern LANs use **switches with point-to-point links** (twisted pair / fibre), so each port is its own collision domain.
- Links run in **full duplex**: separate transmit and receive pairs allow both ends to send simultaneously, so collisions cannot occur and CSMA/CD is disabled.
- Gigabit and faster Ethernet standards (1G, 10G, 100G) in practice only define full-duplex operation.
- CSMA/CD only remains relevant for legacy hubs or half-duplex mismatches, which show up as late collisions and poor performance.

Question 7: CSMA/CA in IEEE 802.11 and comparison with CSMA/CD

Wireless stations cannot reliably detect collisions: a radio cannot transmit and listen at the same time on one channel, and the *hidden terminal problem* means two stations out of each other's range may both transmit to the same AP. 802.11 therefore uses **CSMA/CA** (Collision Avoidance) in the Distributed Coordination Function (DCF):

- **Carrier sense:** the station senses the channel physically and virtually (through the NAV timer).
- **IFS wait:** if idle, the station waits a DIFS interval; if busy, it defers until the channel is free.
- **Random back-off:** it then picks a random back-off counter that decrements only while the channel is idle, reducing the chance that stations that were waiting transmit simultaneously.
- **Transmit and ACK:** after sending, the receiver replies with an ACK after SIFS. No ACK means the sender assumes a collision, doubles its contention window and retries.
- **Optional RTS/CTS:** the sender issues Request To Send and the AP replies Clear To Send, which tells all nodes in range to set their NAV. This solves the hidden-terminal problem for large frames.

Aspect	CSMA/CD (Ethernet, 802.3)	CSMA/CA (Wi-Fi, 802.11)
Objective	Detect a collision and recover fast	Avoid collisions before they happen
Medium	Wired, can detect collisions	Wireless, cannot detect while sending
Acknowledgement	None at the MAC layer	Mandatory ACK for unicast frames
Back-off	After a collision has occurred	Before every transmission and after a failure
Hidden terminal	Not a problem	Handled by RTS/CTS and NAV
Overhead	Low	Higher (IFS, back-off, ACKs)

Question 8: Wireless LAN design for a college laboratory

Scenario: a laboratory of about 12 m x 8 m with 40 workstations and student laptops (approximately 60 concurrent devices).

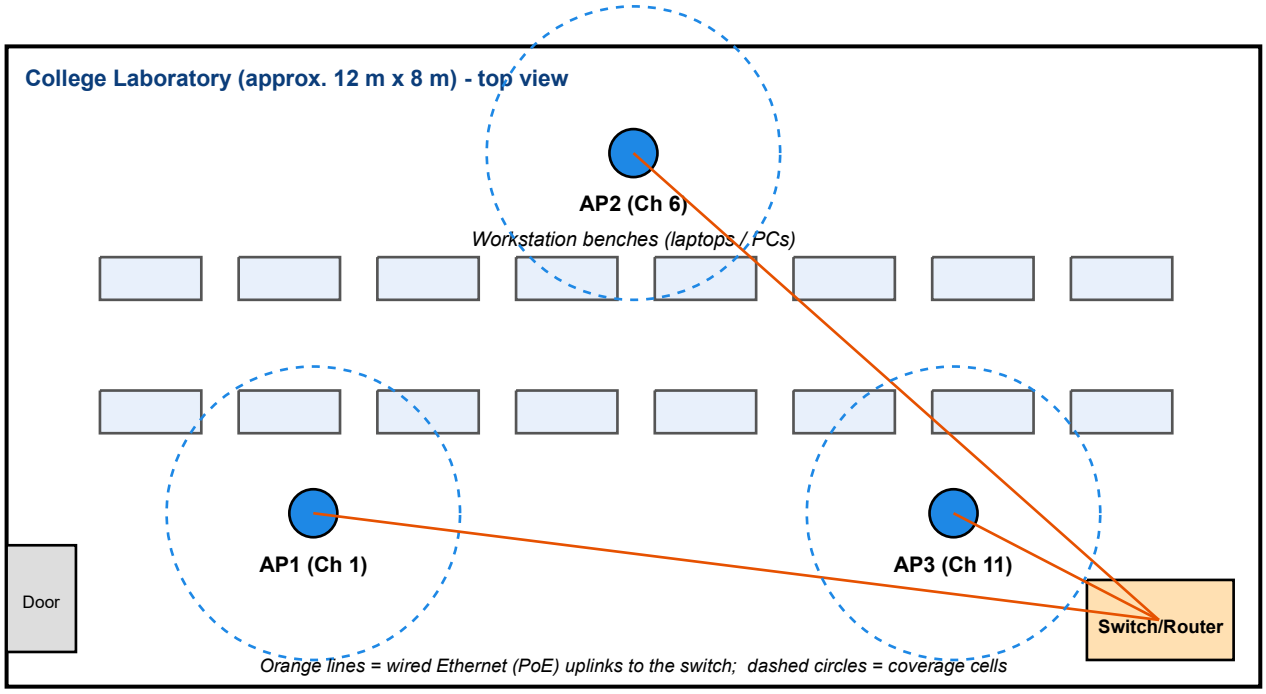


Figure 3: Proposed WLAN layout with three access points on non-overlapping channels.

Design item	Specification
Access point placement	Three ceiling-mounted APs (about 2.7 m high) placed to give roughly 15 - 20% cell overlap so that roaming is seamless; keep them away from metal cabinets and thick walls. Powered by PoE from the switch.
Standard / band	Wi-Fi 5 or Wi-Fi 6 (802.11ac/ax), dual band. Prefer 5 GHz for clients (less interference, more channels); 2.4 GHz for legacy devices.
SSID	**College-Lab** (staff/students, WPA2/WPA3-Enterprise or WPA2/WPA3-Personal) and **College-Guest** (isolated, internet only). Same SSID on all APs to allow roaming; SSID broadcast on.
Channels	2.4 GHz: non-overlapping channels 1, 6 and 11 assigned to AP1, AP2 and AP3 respectively. 5 GHz: use 20/40 MHz-wide channels (e.g. 36, 40, 44) to avoid co-channel interference.
IP addressing	Private subnet 192.168.10.0/24 for the lab (DHCP pool .50 - .250, gateway 192.168.10.1); APs and infrastructure with static IPs (.2 - .10) in a management VLAN, and guest VLAN 192.168.20.0/24.
Wired backbone	APs connect to a PoE-capable Gigabit switch, which uplinks to the router/firewall with NAT to the college network.
Security	WPA3 (or WPA2-AES/CCMP), strong passphrase or 802.1X with RADIUS; disable WPS and WEP; change default admin password; MAC filtering is optional and not a real security measure; client isolation on the guest network; firmware kept up to date.
Capacity	About 20 clients per AP, well within the limits of each radio; rate limits set on the guest VLAN.

Question 9: Mobile IP and the need for mobility support

Problem: an IP address has two roles - identifying a host and indicating its location (the network prefix). When a device moves to another network its address no longer matches the local prefix, so packets are

routed to the old network and the device would need a new address, which breaks ongoing TCP connections. **Mobile IP** (RFC 5944 for IPv4) lets a device keep its permanent address while moving, so applications are not interrupted.

Components:

- **Mobile Node (MN):** the device that moves.
- **Home Agent (HA):** a router on the home network that keeps track of the MN's location and forwards its packets.
- **Foreign Agent (FA):** a router on the visited network that offers a care-of address.
- **Home address:** permanent IP address of the MN. **Care-of Address (CoA):** temporary address that reflects the current location.

Operation:

- **Agent discovery:** HA/FA send advertisements (ICMP router discovery extensions) so the MN knows it has moved.
- **Registration:** the MN registers its CoA with the HA through the FA.
- **Tunnelling:** a correspondent node sends packets to the home address; the HA intercepts them and encapsulates (IP-in-IP) to the CoA; the FA or MN decapsulates them.
- **Reply path:** the MN sends packets directly to the correspondent, or uses reverse tunnelling if ingress filtering blocks it (triangle routing).

Need: smartphones, laptops and vehicles change networks constantly (Wi-Fi to cellular, campus to home). Mobility support keeps VoIP calls, downloads and remote sessions alive. Triangle routing is a drawback that route optimisation (in Mobile IPv6) overcomes.

Question 10: Packet-level comparison of wired Ethernet vs wireless LAN frame

Wireshark can display 802.11 frames only when the adapter is in monitor mode (on Windows, often an external adapter is needed). Where that is not possible, the standard frame layouts below (documented examples) can be compared.

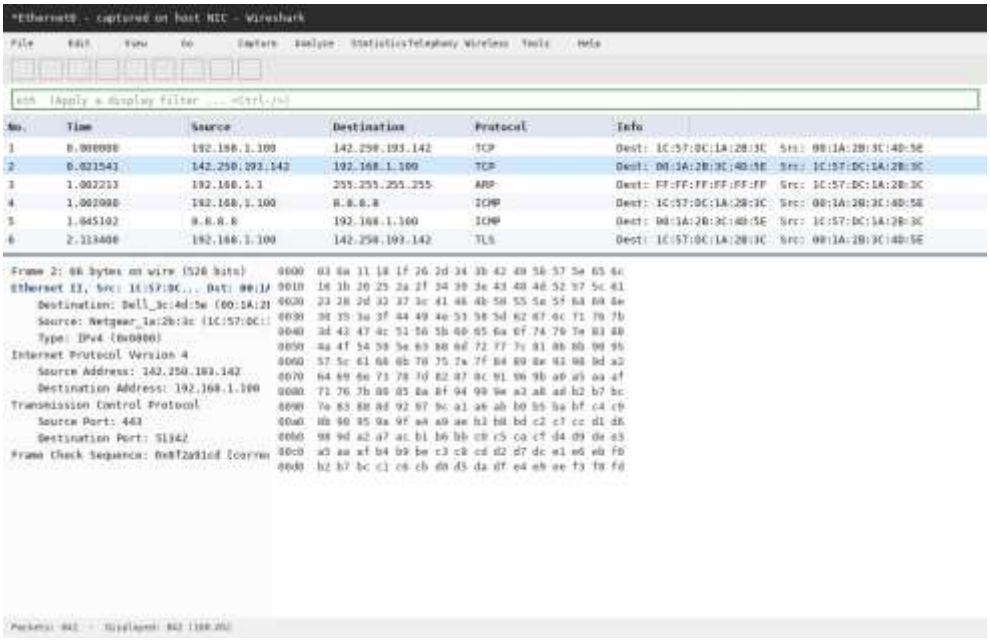
Figure 4a: Wired Ethernet II frame

Preamble 7 B	SFD 1 B	Dest MAC 6 B	Source MAC 6 B	Type/Length 2 B	Payload (Data + Pad) 46 - 1500 B	FCS 4 B

Figure 4b: IEEE 802.11 data frame (generic MAC header)

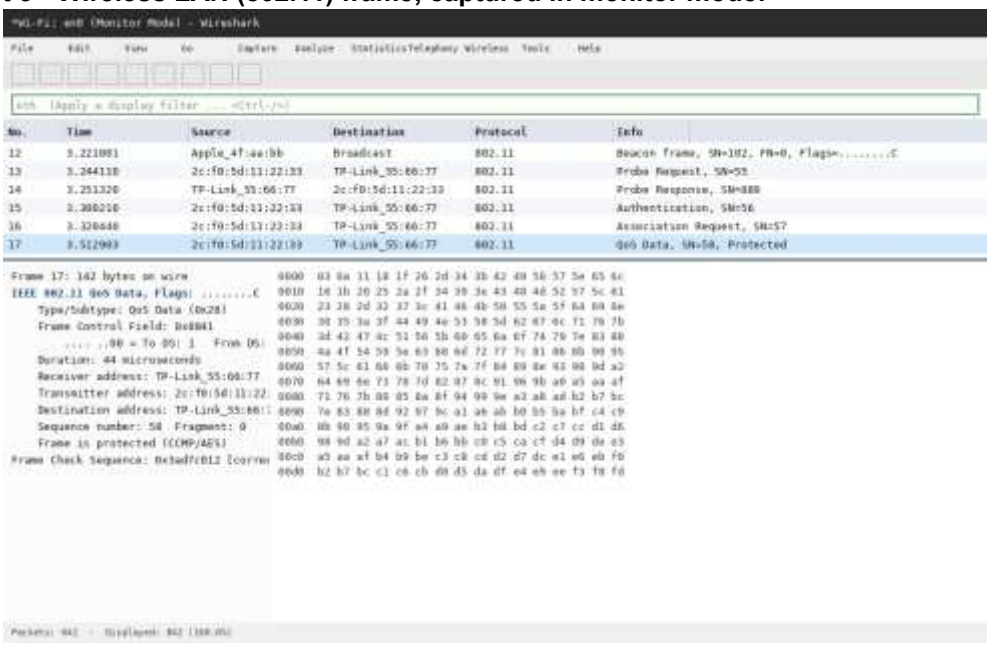
Frame Ctrl 2 B	Dur/ID 2 B	Addr 1 (RA) 6 B	Addr 2 (TA) 6 B	Addr 3 6 B	Seq Ctrl 2 B	Addr 4 6 B (opt)	Frame Body 0 - 2312 B	FCS 4 B

Screenshot 2 - Wired Ethernet frame:



Captured Ethernet II frame: 2 MAC address fields, EtherType 0x0800, no acknowledgement or duration fields.

Screenshot 3 - Wireless LAN (802.11) frame, captured in monitor mode:



Captured 802.11 QoS Data frame: Frame Control, Duration, three/four address fields (Receiver, Transmitter, Destination), Sequence Control and a Protected flag (CCMP/AES encryption) that a wired Ethernet frame does not carry.

Feature	Wired Ethernet II	Wireless LAN (802.11)
Header size	14 bytes (excluding preamble)	24 bytes (30 with Addr 4; +2 for QoS)
Address fields	2 (destination, source)	Up to 4 (receiver, transmitter, and destination/source depending on To-DS/From-DS bits)
Frame types	Data only at the MAC level	Management (beacon, probe, auth), Control (RTS, CTS, ACK) and Data frames

Control information	EtherType only	Frame Control field: version, type/subtype, To DS/From DS, retry, power management, protected (encrypted) flag
Feature	**Wired Ethernet II**	**Wireless LAN (802.11)**
Acknowledgement	None	Link-layer ACK for every unicast frame
Duration / NAV	Not present	Duration/ID field used for virtual carrier sensing
Sequence number	Not present	Sequence Control field for duplicate detection and fragmentation
Max payload	1500 B	2304 - 2312 B
Security	None at Layer 2 (802.1X, MACsec optional)	Encryption at Layer 2 (WPA2 CCMP / WPA3 GCMP) with the Protected bit set
FCS	4 B CRC-32	4 B CRC-32

Conclusion: the wireless frame is more complex because the medium is unreliable and shared, requiring extra addressing, control, acknowledgement and security fields that a wired switched LAN does not need.

Deliverables Checklist

Deliverable	Where in report
Ethernet frame diagram	Q2 (Figure 1)
Wireshark screenshots	Q3 and Q10
Wireless LAN design	Q8 (Figure 3 and table)
Comparison table	Q5 (also Q7 and Q10)
README.md	Included in the GitHub repository

References

- Forouzan, B. A., *Data Communications and Networking*, McGraw-Hill.
- Tanenbaum, A. S. and Wetherall, D., *Computer Networks*, Pearson.
- IEEE 802.3 (Ethernet) and IEEE 802.11 (Wireless LAN) standards.
- RFC 5944 - IP Mobility Support for IPv4, Revised.
- Wireshark User's Guide, https://www.wireshark.org/docs/